

Supplemental Materials:

Supplementary Methods: Derivation of WHO fetal growth chart–based percentile variables

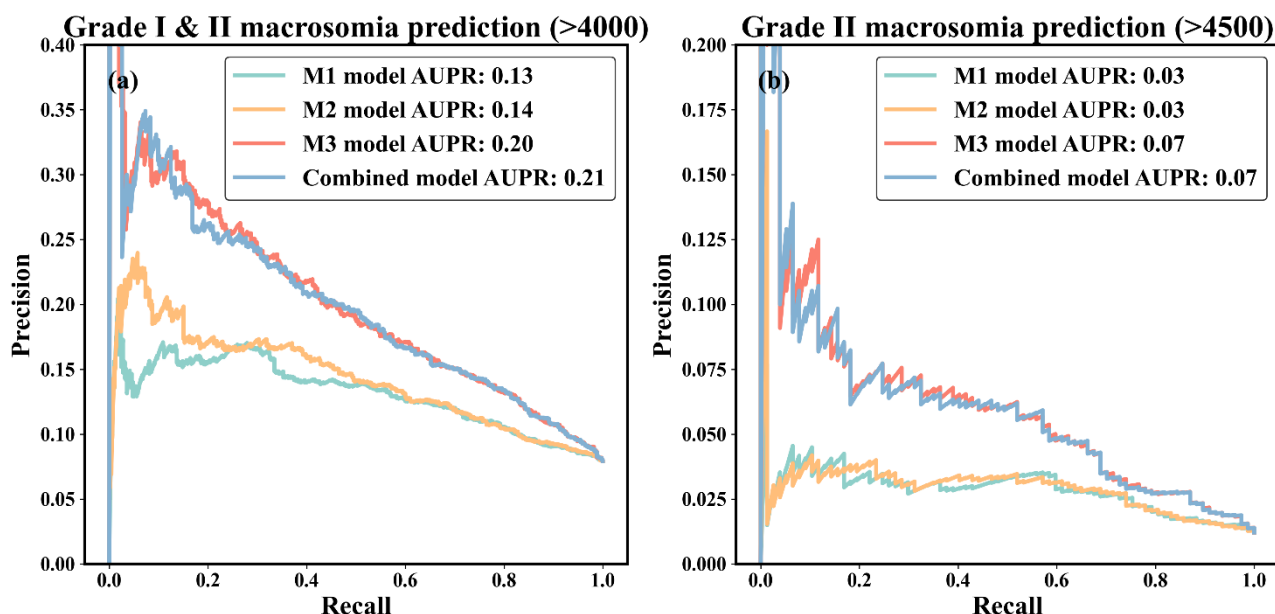
For each pregnancy with an available ultrasound examination at Visit 2 or Visit 3, gestational age at scan and one fetal biometric measurement were mapped to the World Health Organization fetal growth chart to obtain a gestational-age-specific percentile. The biometric inputs included head circumference, biparietal diameter, abdominal circumference, femur length, and estimated fetal weight. Estimated fetal weight was first calculated from ultrasound biometry using the Hadlock formula and then converted into a WHO chart–based percentile.

Percentiles were computed separately for Visit 2 and Visit 3. Accordingly, five WHO-derived percentile variables were generated at Visit 2 and five additional variables were generated at Visit 3. Variable names followed the format WHO-[biometric variable]-[visit], where the suffix “2” denotes Visit 2 and the suffix “3” denotes Visit 3. For example, WHO-EFW-2 represents the percentile derived from estimated fetal weight and gestational age at Visit 2, whereas WHO-AC-3 represents the percentile derived from abdominal circumference and gestational age at Visit 3.

All WHO-derived percentile variables were evaluated as ultrasound-based benchmark predictors in the comparison with the logistic regression models. Their discriminative performance was summarized as area under the receiver operating characteristic curve values in Table 2 of the main manuscript. In addition, mean AUC values across the five WHO-derived percentiles at each visit were used for statistical comparison with the M1, M2, and M3 logistic regression models using the Mann–Whitney U test across repeated cross-validation runs, as described in the main Methods section. The estimated fetal weight percentile was additionally emphasized in the narrative interpretation because it most closely reflects routine clinical assessment of fetal overgrowth.

Figure S1. Precision–recall (PR) curves for macrosomia prediction.

(a) >4000 g (prevalence 8%). (b) >4500 g (prevalence 1%). Curves were computed from pooled out-of-fold test predictions under stratified 10-fold cross-validation and summarized over 1000 repeated runs.



Models: M1 (≤ 13 weeks), M2 (M1+V2 ultrasound), M3 (M1+V3 ultrasound), Combined (≤ 30 weeks).
AUPRC (area under the precision–recall curve) values by model: >4000 g—M1 0.13, M2 0.14, M3 0.20, Combined 0.21; >4500 g—M1 0.03, M2 0.03, M3 0.07, Combined 0.07.

Figure S2. Calibration plots for macrosomia prediction.

(a) >4000 g. (b) >4500 g. Reliability curves use isotonic smoothing on pooled out-of-fold test predictions. Models: M1 (≤ 13 weeks), M2 (M1+V2 ultrasound), M3 (M1+V3 ultrasound), Combined (≤ 30 weeks). Brier scores(a): M1 0.072, M2 0.071, M3 0.069, Combined 0.068. Brier scores(b): M1 0.012, M2 0.012, M3 0.012, Combined 0.011.

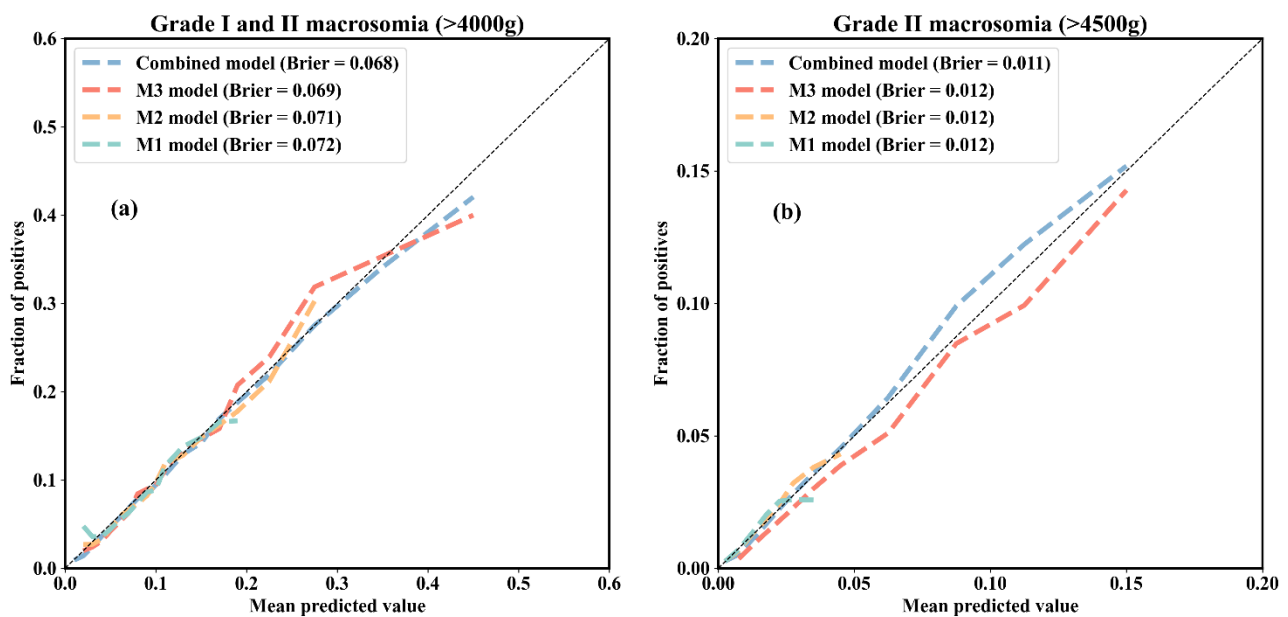
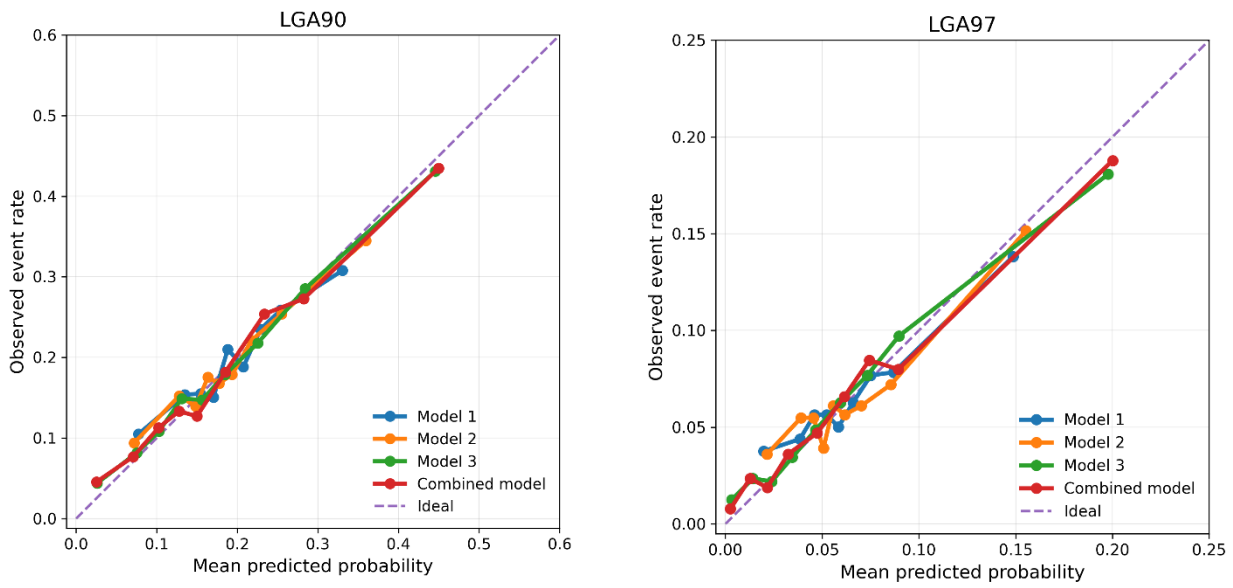
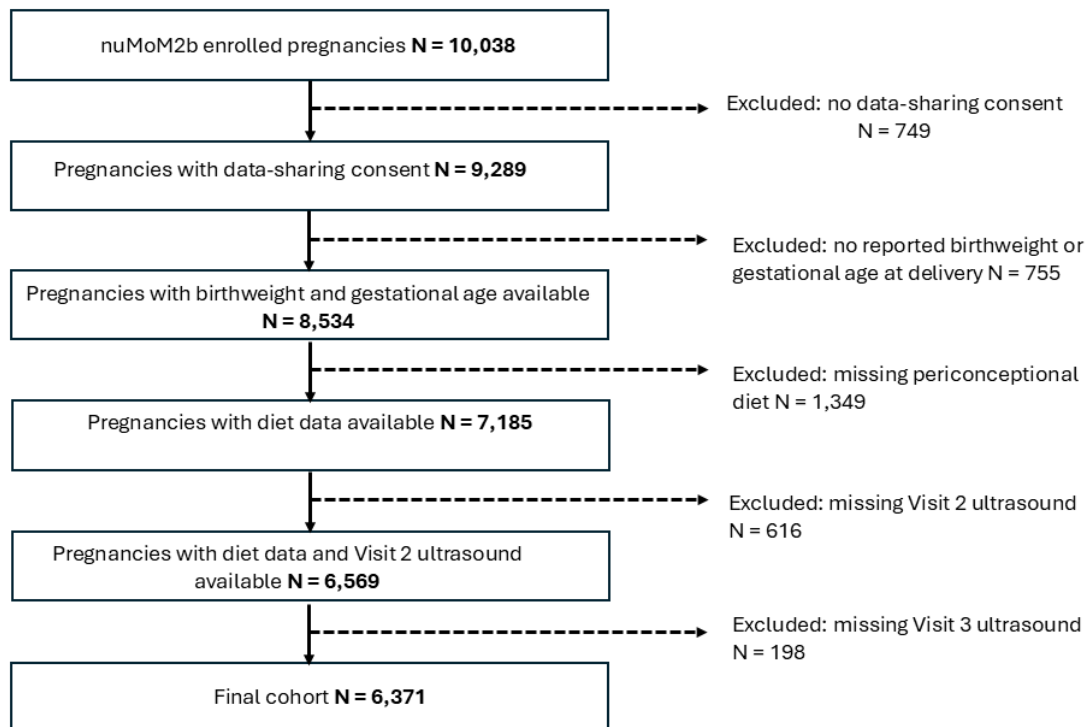


Figure S3. Calibration plots for LGA sensitivity outcomes. Calibration plots were generated from calibrated out-of-fold predicted probabilities from repeated 10-fold cross-validation. Each point represents one predicted-risk block, plotted as the mean predicted probability against the observed event rate. The dashed diagonal line indicates perfect calibration. Panels show sensitivity outcomes defined as birthweight above the INTERGROWTH-21st gestational-age- and sex-specific (a) 90th percentile and (b) 97th percentile. Model 1, Model 2, Model 3, and the Combined model are shown separately.



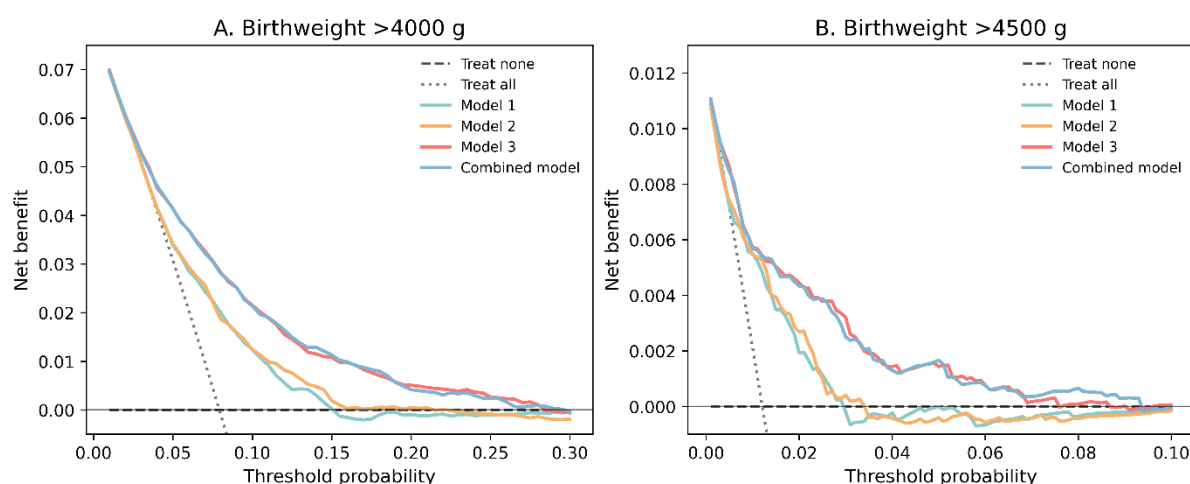
Supplementary Figure S4. Participant flow for derivation of the analytic cohort.



Note: Exclusions are shown sequentially and are therefore mutually exclusive. Among 10,038 women enrolled in the nuMoM2b cohort, 749 were excluded because data-sharing consent was unavailable.

Among the remaining 9,289 pregnancies, 755 were excluded because birthweight or gestational age at delivery was not reported, leaving 8,534 pregnancies with outcome information. Of these, 1,349 were excluded because of missing periconceptional diet data. Among the remaining 7,185 pregnancies, 616 lacked Visit 2 ultrasound data, including 523 missing Visit 2 only and 93 missing both Visit 2 and Visit 3. Among the 6,569 pregnancies with Visit 2 ultrasound available, 198 additionally lacked Visit 3 ultrasound data. The final analytic cohort included 6,371 pregnancies.

Supplementary Figure S5. Decision curve analysis for primary macrosomia outcomes.



Decision curve analysis was performed using calibrated out-of-fold predicted probabilities from the internally validated logistic-regression models. Net benefit was calculated across clinically relevant threshold-probability ranges and compared with treat-all and treat-none reference strategies. Panels show decision curves for (a) birthweight >4000 g and (b) birthweight >4500 g. Because birthweight >4500 g was rare, a lower threshold-probability range was used for this outcome. The curves are intended to illustrate potential clinical utility across risk-threshold ranges and should not be interpreted as defining recommended intervention thresholds.

Table S1. Univariate associations for macrosomia >4000 g.

For each predictor: test used (t-test with Welch as needed, Mann–Whitney U, or χ^2 /Fisher), effect size (Cohen’s d/Cliff’s delta/odds ratio), raw p, and BH-FDR q ($q < 0.05$, two-sided $\alpha = 0.05$). Variable names/units follow Table 1.

Variable	Type	Test	p_value	Effect_size	Effect_type	q_value	Significant
Age	numeric	Mann–Whitney U	<0.001	-0.105	Cliff's delta	<0.001	TRUE
Non-Hispanic white	categorical	Chi-square	<0.001	0.045	Cramér's V	<0.001	TRUE
Non-Hispanic black	categorical	Chi-square	<0.05	0.0322	Cramér's V	<0.05	TRUE

Hispanic	categorical	Chi-square	0.25	0.014	Cramér's V	0.30	FALSE
Asian and Rest	categorical	Chi-square	0.06	0.0235	Cramér's V	0.08	FALSE
Income	numeric	Mann-Whitney U	<0.001	-0.105	Cliff's delta	<0.001	TRUE
Smoke	categorical	Chi-square	0.75	0.004	Cramér's V	0.77	FALSE
Fetal sex	categorical	Chi-square	<0.001	0.089	Cramér's V	<0.001	TRUE
Pre-pregnancy BMI	numeric	Mann-Whitney U	<0.001	-0.189	Cliff's delta	<0.001	TRUE
Preexisting diabetes	categorical	Chi-square	0.42	0.010	Cramér's V	0.46	FALSE
HEI - Total Vegetables	numeric	Mann-Whitney U	<0.05	-0.065	Cliff's delta	<0.05	TRUE
HEI - Vegetables & Legumes	numeric	Mann-Whitney U	0.26	-0.028	Cliff's delta	0.31	FALSE
HEI - Total Fruit	numeric	Mann-Whitney U	<0.05	-0.080	Cliff's delta	<0.05	TRUE
HEI - Whole Fruit	numeric	Mann-Whitney U	<0.001	-0.094	Cliff's delta	<0.001	TRUE
HEI - Whole Grains	numeric	Mann-Whitney U	<0.05	-0.065	Cliff's delta	<0.05	TRUE
HEI - Milk	numeric	Mann-Whitney U	0.10	-0.043	Cliff's delta	0.13	FALSE
HEI - Meat and Beans	numeric	Mann-Whitney U	0.31	-0.026	Cliff's delta	0.36	FALSE
HEI - Seafood & Plant Protein	numeric	Mann-Whitney U	<0.05	-0.059	Cliff's delta	<0.05	TRUE
HEI - Fatty Acid Ratio	numeric	Mann-Whitney U	0.79	0.007	Cliff's delta	0.80	FALSE
HEI - Sodium	numeric	Mann-Whitney U	0.07	0.047	Cliff's delta	0.09	FALSE
HEI - Refined Grains	numeric	Mann-Whitney U	<0.05	-0.063	Cliff's delta	<0.05	TRUE
HEI - SoFAAS Cals	numeric	Mann-Whitney U	<0.05	-0.073	Cliff's delta	<0.05	TRUE
HEI - Total Score	numeric	Mann-Whitney U	<0.001	-0.095	Cliff's delta	<0.001	TRUE

AHEI - Vegetable servings score	numeric	Mann-Whitney U	<0.05	-0.065	Cliff's delta	<0.05	TRUE
AHEI - Fruit servings score	numeric	Mann-Whitney U	<0.001	-0.096	Cliff's delta	<0.001	TRUE
AHEI - Whole grain servings score	numeric	Mann-Whitney U	<0.05	-0.064	Cliff's delta	<0.05	TRUE
AHEI - Sugary beverages servings score	numeric	Mann-Whitney U	0.15	-0.036	Cliff's delta	0.18	FALSE
AHEI - Nuts and legumes servings score	numeric	Mann-Whitney U	<0.05	-0.056	Cliff's delta	0.05	FALSE
AHEI - Red meats servings score	numeric	Mann-Whitney U	0.57	0.015	Cliff's delta	0.60	FALSE
AHEI - Trans-fat percent score	numeric	Mann-Whitney U	0.23	-0.032	Cliff's delta	0.27	FALSE
AHEI - DHA & EPA intake score	numeric	Mann-Whitney U	0.98	0.001	Cliff's delta	0.98	FALSE
AHEI - Polyunsaturated fat percent score	numeric	Mann-Whitney U	0.54	-0.016	Cliff's delta	0.58	FALSE
AHEI - Sodium intake score	numeric	Mann-Whitney U	0.37	0.023	Cliff's delta	0.42	FALSE
AHEI - Alcoholic drinks score	categorical	Chi-square	<0.001	0.057	Cramér's V	<0.001	TRUE
AHEI - Total Score	numeric	Mann-Whitney U	<0.001	-0.093	Cliff's delta	<0.05	TRUE
BMI at Visit 1	numeric	Mann-Whitney U	<0.001	-0.207	Cliff's delta	<0.001	TRUE
waist circumference at Visit 1	numeric	Mann-Whitney U	<0.001	-0.098	Cliff's delta	<0.001	TRUE
waist over iliac crest at Visit 1	numeric	Mann-Whitney U	<0.001	-0.197	Cliff's delta	<0.001	TRUE
Hip circumference at Visit 1	numeric	Mann-Whitney U	<0.001	-0.202	Cliff's delta	<0.001	TRUE
Neck circumference at Visit 1	numeric	Mann-Whitney U	<0.001	-0.206	Cliff's delta	<0.001	TRUE
SBP at Visit 1	numeric	Mann-Whitney U	<0.001	-0.233	Cliff's delta	<0.001	TRUE

DBP at Visit 1	numeric	Mann-Whitney U	<0.05	-0.053	Cliff's delta	0.06	FALSE
GWG at Visit 1	numeric	Mann-Whitney U	<0.05	-0.060	Cliff's delta	<0.05	TRUE
GWG at Visit 2	numeric	Mann-Whitney U	<0.001	-0.147	Cliff's delta	<0.001	TRUE
SBP at Visit 2	numeric	Mann-Whitney U	<0.05	-0.070	Cliff's delta	<0.05	TRUE
DBP at Visit 2	numeric	Mann-Whitney U	0.06	-0.049	Cliff's delta	0.08	FALSE
GDM	categorical	Chi-square	0.05	0.024	Cramér's V	0.07	FALSE
GA at Visit 2	numeric	Mann-Whitney U	<0.05	-0.052	Cliff's delta	0.06	FALSE
BPD at Visit 2	numeric	Mann-Whitney U	<0.001	-0.147	Cliff's delta	<0.001	TRUE
HC at Visit 2	numeric	Mann-Whitney U	<0.001	-0.149	Cliff's delta	<0.001	TRUE
AC at Visit 2	numeric	Mann-Whitney U	<0.001	-0.175	Cliff's delta	<0.001	TRUE
FL at Visit 2	numeric	Mann-Whitney U	<0.001	-0.109	Cliff's delta	<0.001	TRUE
EFW at Visit 2	numeric	Mann-Whitney U	<0.001	-0.152	Cliff's delta	<0.001	TRUE
GWG at Visit 3	numeric	Mann-Whitney U	<0.001	-0.183	Cliff's delta	<0.001	TRUE
SBP at Visit 3	numeric	Mann-Whitney U	<0.05	-0.053	Cliff's delta	0.06	FALSE
DBP at Visit 3	numeric	Mann-Whitney U	0.44	-0.020	Cliff's delta	0.48	FALSE
GA at Visit 3	numeric	Mann-Whitney U	<0.001	-0.091	Cliff's delta	0.001	TRUE
BPD at Visit 3	numeric	Mann-Whitney U	<0.001	-0.301	Cliff's delta	<0.001	TRUE
HC at Visit 3	numeric	Mann-Whitney U	<0.001	-0.288	Cliff's delta	<0.001	TRUE
AC at Visit 3	numeric	Mann-Whitney U	<0.001	-0.362	Cliff's delta	<0.001	TRUE

FL at Visit 3	numeric	Mann–Whitney U	<0.001	-0.230	Cliff's delta	<0.001	TRUE
EFW at Visit 3	numeric	Mann–Whitney U	<0.001	-0.103	Cliff's delta	<0.001	TRUE
AFI - Quadrant 1 at Visit 3	numeric	Mann–Whitney U	<0.001	-0.131	Cliff's delta	<0.001	TRUE
AFI - Quadrant 2 at Visit 3	numeric	Mann–Whitney U	<0.05	-0.070	Cliff's delta	<0.05	TRUE
AFI - Quadrant 3 at Visit 3	numeric	Mann–Whitney U	<0.001	-0.335	Cliff's delta	<0.001	TRUE
AFI - Quadrant 4 at Visit 3	numeric	Mann–Whitney U	<0.001	-0.381	Cliff's delta	<0.001	TRUE

Table S2. Univariate associations for macrosomia >4500 g.

For each predictor: test used (t-test with Welch as needed, Mann–Whitney U, or χ^2 /Fisher), effect size (Cohen's *d*/Cliff's delta/odds ratio), raw *p*, and BH-FDR *q* (*q*<0.05, two-sided α =0.05). Variable names/units follow Table 1.

Variable	Type	Test	P value	Effect size	Effect type	Q value	Significant
Age	numeric	Mann–Whitney U	<0.001	-0.202	Cliff's delta	<0.05	TRUE
Non-Hispanic white	categorical	Chi-square	0.68	0.005	Cramér's V	0.74	FALSE
Non-Hispanic black	categorical	Chi-square	0.10	0.020	Cramér's V	0.19	FALSE
Hispanic	categorical	Chi-square	0.14	0.018	Cramér's V	0.23	FALSE
Asian and Rest	categorical	Chi-square	0.43	0.009	Cramér's V	0.51	FALSE
Income	numeric	Mann–Whitney U	<0.001	-0.213	Cliff's delta	<0.05	TRUE
Smoke	categorical	Chi-square	0.21	0.015	Cramér's V	0.29	FALSE
Fetal sex	categorical	Chi-square	<0.001	0.045	Cramér's V	<0.05	TRUE
Pre-pregnancy BMI	numeric	Mann–Whitney U	<0.001	-0.261	Cliff's delta	<0.001	TRUE
Preexisting diabetes	categorical	Fisher's exact (2x2)	0.28	0.011	Cramér's V	0.37	FALSE

HEI - Total Vegetables	numeric	Mann-Whitney U	0.12	-0.098	Cliff's delta	0.22	FALSE
HEI - Vegetables & Legumes	numeric	Mann-Whitney U	0.66	-0.027	Cliff's delta	0.72	FALSE
HEI - Total Fruit	numeric	Mann-Whitney U	0.24	-0.073	Cliff's delta	0.32	FALSE
HEI - Whole Fruit	numeric	Mann-Whitney U	0.16	-0.083	Cliff's delta	0.25	FALSE
HEI - Whole Grains	numeric	Mann-Whitney U	0.40	-0.055	Cliff's delta	0.48	FALSE
HEI - Milk	numeric	Mann-Whitney U	0.45	0.049	Cliff's delta	0.52	FALSE
HEI - Meat and Beans	numeric	Mann-Whitney U	0.13	-0.094	Cliff's delta	0.22	FALSE
HEI - Seafood & Plant Protein	numeric	Mann-Whitney U	0.05	-0.120	Cliff's delta	0.12	FALSE
HEI - Fatty Acid Ratio	numeric	Mann-Whitney U	<0.05	-0.146	Cliff's delta	0.06	FALSE
HEI - Sodium	numeric	Mann-Whitney U	<0.05	0.142	Cliff's delta	0.07	FALSE
HEI - Refined Grains	numeric	Mann-Whitney U	0.78	0.017	Cliff's delta	0.83	FALSE
HEI - SoFAAS Cals	numeric	Mann-Whitney U	<0.05	-0.172	Cliff's delta	<0.05	TRUE
HEI - Total Score	numeric	Mann-Whitney U	0.06	-0.124	Cliff's delta	0.12	FALSE
AHEI - Vegetable servings score	numeric	Mann-Whitney U	0.47	-0.047	Cliff's delta	0.53	FALSE
AHEI - Fruit servings score	numeric	Mann-Whitney U	0.18	-0.088	Cliff's delta	0.26	FALSE
AHEI - Whole grain servings score	numeric	Mann-Whitney U	0.64	-0.035	Cliff's delta	0.71	FALSE
AHEI - Sugary beverages servings score	numeric	Mann-Whitney U	0.07	-0.112	Cliff's delta	0.13	FALSE
AHEI - Nuts and legumes servings score	numeric	Mann-Whitney U	0.21	-0.081	Cliff's delta	0.29	FALSE

AHEI - Red meats servings score	numeric	Mann-Whitney U	0.96	0.003	Cliff's delta	0.96	FALSE
AHEI - Trans-fat percent score	numeric	Mann-Whitney U	0.34	-0.062	Cliff's delta	0.43	FALSE
AHEI - DHA & EPA intake score	numeric	Mann-Whitney U	0.96	0.003	Cliff's delta	0.96	FALSE
AHEI - Polyunsaturated fat percent score	numeric	Mann-Whitney U	<0.05	-0.135	Cliff's delta	0.08	FALSE
AHEI - Sodium intake score	numeric	Mann-Whitney U	0.86	-0.011	Cliff's delta	0.88	FALSE
AHEI - Alcoholic drinks score	categorical	Chi-square	<0.001	0.056	Cramér's V	<0.001	TRUE
AHEI - Total Score	numeric	Mann-Whitney U	<0.001	-0.195	Cliff's delta	<0.05	TRUE
BMI at visit 1	numeric	Mann-Whitney U	<0.001	-0.289	Cliff's delta	<0.001	TRUE
waist circumference at visit 1	numeric	Mann-Whitney U	0.05	-0.128	Cliff's delta	0.11	FALSE
waist over iliac crest at visit 1	numeric	Mann-Whitney U	<0.001	-0.297	Cliff's delta	<0.001	TRUE
Hip circumference at visit 1	numeric	Mann-Whitney U	<0.001	-0.314	Cliff's delta	<0.001	TRUE
Neck circumference at visit 1	numeric	Mann-Whitney U	<0.001	-0.315	Cliff's delta	<0.001	TRUE
SBP at visit 1	numeric	Mann-Whitney U	<0.001	-0.316	Cliff's delta	<0.001	TRUE
DBP at visit 1	numeric	Mann-Whitney U	<0.05	-0.177	Cliff's delta	<0.05	TRUE
GWG at visit 1	numeric	Mann-Whitney U	0.15	-0.093	Cliff's delta	0.24	FALSE
GWG at visit 2	numeric	Mann-Whitney U	<0.05	-0.209	Cliff's delta	<0.05	TRUE
SBP at visit 2	numeric	Mann-Whitney U	0.14	-0.096	Cliff's delta	0.23	FALSE
DBP at visit 2	numeric	Mann-Whitney U	0.18	-0.088	Cliff's delta	0.26	FALSE

GDM	categorical	Fisher's exact (2x2)	0.39	0.010	Cramér's V	0.48	FALSE
GA at visit 2	numeric	Mann-Whitney U	0.79	-0.016	Cliff's delta	0.83	FALSE
BPD at visit 2	numeric	Mann-Whitney U	0.08	-0.112	Cliff's delta	0.16	FALSE
HC at visit 2	numeric	Mann-Whitney U	<0.05	-0.154	Cliff's delta	0.05	FALSE
AC at visit 2	numeric	Mann-Whitney U	<0.05	-0.198	Cliff's delta	<0.05	TRUE
FL at visit 2	numeric	Mann-Whitney U	0.17	-0.088	Cliff's delta	0.26	FALSE
EFW at visit 2	numeric	Mann-Whitney U	<0.05	-0.150	Cliff's delta	0.05	FALSE
GWG at visit 3	numeric	Mann-Whitney U	<0.05	-0.203	Cliff's delta	<0.05	TRUE
SBP at visit 3	numeric	Mann-Whitney U	0.10	-0.106	Cliff's delta	0.19	FALSE
DBP at visit 3	numeric	Mann-Whitney U	0.34	-0.061	Cliff's delta	0.43	FALSE
GA at visit 3	numeric	Mann-Whitney U	0.23	-0.079	Cliff's delta	0.31	FALSE
BPD at visit 3	numeric	Mann-Whitney U	<0.001	-0.431	Cliff's delta	<0.001	TRUE
HC at visit 3	numeric	Mann-Whitney U	<0.001	-0.411	Cliff's delta	<0.001	TRUE
AC at visit 3	numeric	Mann-Whitney U	<0.001	-0.501	Cliff's delta	<0.001	TRUE
FL at visit 3	numeric	Mann-Whitney U	<0.001	-0.330	Cliff's delta	<0.001	TRUE
EFW at visit 3	numeric	Mann-Whitney U	<0.05	-0.169	Cliff's delta	<0.05	TRUE
AFI - Quadrant 1 at visit 3	numeric	Mann-Whitney U	<0.001	-0.305	Cliff's delta	<0.001	TRUE
AFI - Quadrant 2 at visit 3	numeric	Mann-Whitney U	<0.05	-0.202	Cliff's delta	<0.05	TRUE

AFI - Quadrant 3 at visit 3	numeric	Mann-Whitney U	<0.001	-0.473	Cliff's delta	<0.001	TRUE
AFI - Quadrant 4 at visit 3	numeric	Mann-Whitney U	<0.001	-0.550	Cliff's delta	<0.001	TRUE

Supplementary Table S3. Definition and analytical role of WHO fetal growth chart-based percentile variables.

Variable name	Visit	nuMoM2b input(s)	WHO chart input	Derived variable meaning	Role in analysis
WHO-HC-2	Visit 2	Head circumference at Visit 2 + gestational age at scan	HC + GA	WHO fetal growth percentile based on head circumference at Visit 2	Included in Table 2 benchmark comparison
WHO-BPD-2	Visit 2	Biparietal diameter at Visit 2 + gestational age at scan	BPD + GA	WHO fetal growth percentile based on biparietal diameter at Visit 2	Included in Table 2 benchmark comparison
WHO-AC-2	Visit 2	Abdominal circumference at Visit 2 + gestational age at scan	AC + GA	WHO fetal growth percentile based on abdominal circumference at Visit 2	Included in Table 2 benchmark comparison
WHO-FL-2	Visit 2	Femur length at Visit 2 + gestational age at scan	FL + GA	WHO fetal growth percentile based on femur length at Visit 2	Included in Table 2 benchmark comparison
WHO-EFW-2	Visit 2	Hadlock-derived estimated fetal weight at Visit 2 + gestational age at scan	EFW + GA	WHO fetal growth percentile based on estimated fetal weight at Visit 2	Included in Table 2 benchmark comparison; clinically emphasized proxy
WHO-HC-3	Visit 3	Head circumference at Visit 3 + gestational age at scan	HC + GA	WHO fetal growth percentile based on head circumference at Visit 3	Included in Table 2 benchmark comparison
WHO-BPD-3	Visit 3	Biparietal diameter at Visit 3 + gestational age at scan	BPD + GA	WHO fetal growth percentile based on biparietal diameter at Visit 3	Included in Table 2 benchmark comparison
WHO-AC-3	Visit 3	Abdominal circumference at Visit 3 + gestational age at scan	AC + GA	WHO fetal growth percentile based on abdominal circumference at Visit 3	Included in Table 2 benchmark comparison
WHO-FL-3	Visit 3	Femur length at Visit 3 + gestational age at scan	FL + GA	WHO fetal growth percentile based on femur length at Visit 3	Included in Table 2 benchmark comparison
WHO-EFW-3	Visit 3	Hadlock-derived estimated fetal	EFW + GA	WHO fetal growth percentile based on	Included in Table 2 benchmark

		weight at Visit 3 + gestational age at scan		estimated fetal weight at Visit 3	comparison; clinically emphasized proxy
--	--	---	--	--------------------------------------	---

Abbreviations: AC, abdominal circumference; BPD, biparietal diameter; EFW, estimated fetal weight; FL, femur length; GA, gestational age; HC, head circumference; WHO, World Health Organization.

Note: Estimated fetal weight was calculated using the Hadlock formula, and WHO growth-chart percentiles were assigned according to gestational age and the corresponding ultrasound-derived fetal biometric measurement. All 10 WHO-derived percentile variables were evaluated in the main benchmark comparison reported in Table 2. Average AUC values across the five WHO-derived variables at each visit were additionally compared with the AUC distributions of M1, M2, and M3, as indicated in the Table 2 footnotes.

Supplementary Table S4. Discrimination and calibration of class-weighted isotonic-calibrated models for LGA sensitivity outcomes.

Outcome	Model	Event rate	AUROC (95% CI)	AUPRC (95% CI)	Brier score (95% CI)	Calibration intercept (95% CI)	Calibration slope (95% CI)
INTERGROWTH-21st birthweight centile >90th percentile	Model 1	20.7%	0.613 (0.596–0.629)	0.279 (0.260–0.300)	0.161 (0.155–0.167)	-0.365 (-0.540 to -0.190)	0.718 (0.593–0.841)
INTERGROWTH-21st birthweight centile >90th percentile	Model 2	20.7%	0.635 (0.618–0.652)	0.311 (0.289–0.335)	0.158 (0.152–0.164)	-0.276 (-0.461 to -0.103)	0.784 (0.649–0.911)
INTERGROWTH-21st birthweight centile >90th percentile	Model 3	20.7%	0.722 (0.707–0.737)	0.412 (0.386–0.441)	0.146 (0.141–0.152)	-0.160 (-0.285 to -0.045)	0.864 (0.774–0.949)
INTERGROWTH-21st birthweight centile >90th percentile	Combined model	20.7%	0.727 (0.712–0.742)	0.412 (0.385–0.439)	0.146 (0.140–0.151)	-0.149 (-0.264 to -0.029)	0.872 (0.786–0.960)
INTERGROWTH-21st birthweight centile >97th percentile	Model 1	7.4%	0.635 (0.608–0.662)	0.127 (0.110–0.149)	0.067 (0.062–0.073)	-0.623 (-1.016 to -0.247)	0.741 (0.585–0.893)
INTERGROWTH-21st birthweight centile >97th percentile	Model 2	7.4%	0.642 (0.614–0.668)	0.139 (0.119–0.166)	0.067 (0.062–0.072)	-0.554 (-0.931 to -0.207)	0.770 (0.629–0.910)
INTERGROWTH-21st birthweight centile >97th percentile	Model 3	7.4%	0.746 (0.723–0.769)	0.215 (0.186–0.252)	0.064 (0.059–0.068)	-0.384 (-0.681 to -0.088)	0.824 (0.706–0.947)
INTERGROWTH-21st birthweight centile >97th percentile	Combined model	7.4%	0.751 (0.728–0.774)	0.218 (0.187–0.254)	0.063 (0.059–0.068)	-0.367 (-0.654 to -0.109)	0.832 (0.713–0.942)

Table note: LGA sensitivity outcomes were defined using INTERGROWTH-21st birthweight centiles according to gestational age at delivery and neonatal sex. Values are based on calibrated out-of-fold predicted probabilities from repeated 10-fold cross-validation. Class weights were defined using the square root of the negative-to-positive sample ratio, with the negative class assigned a weight of 1. Probability calibration was performed using isotonic regression within the cross-validation framework. Confidence intervals were obtained by bootstrap resampling. AUROC: area under the receiver operating characteristic curve; AUPRC: area under the precision–recall curve; CI: confidence interval; LGA: large for gestational age.

Supplementary Table S5. Discrimination and calibration of class-weighted isotonic-calibrated models for the primary macrosomia outcomes.

Outcome	Model	Event rate	AUROC (95% CI)	AUPRC (95% CI)	Brier score (95% CI)	Calibration intercept (95% CI)	Calibration slope (95% CI)
Birthweight >4000 g	Model 1	7.9%	0.659 (0.635–0.674)	0.132 (0.116–0.152)	0.072 (0.066–0.077)	-0.761 (-1.258 to -0.291)	0.672 (0.467–0.866)
Birthweight >4000 g	Model 2	7.9%	0.668 (0.646–0.684)	0.136 (0.119–0.157)	0.071 (0.067–0.077)	-0.676 (-1.124 to -0.282)	0.709 (0.529–0.878)
Birthweight >4000 g	Model 3	7.9%	0.745 (0.723–0.766)	0.203 (0.187–0.235)	0.069 (0.064–0.074)	-0.373 (-0.665 to -0.111)	0.826 (0.708–0.942)
Birthweight >4000 g	Combined model	7.9%	0.749 (0.726–0.781)	0.211 (0.190–0.236)	0.068 (0.064–0.074)	-0.344 (-0.606 to -0.076)	0.839 (0.724–0.958)
Birthweight >4500 g	Model 1	1.2%	0.735 (0.688–0.777)	0.028 (0.020–0.042)	0.012 (0.009–0.014)	-2.083 (-3.359 to -0.648)	0.485 (0.214–0.833)
Birthweight >4500 g	Model 2	1.2%	0.745 (0.689–0.783)	0.030 (0.021–0.036)	0.012 (0.009–0.015)	-1.980 (-3.088 to -0.801)	0.519 (0.279–0.800)
Birthweight >4500 g	Model 3	1.2%	0.822 (0.790–0.862)	0.066 (0.041–0.119)	0.012 (0.009–0.014)	-1.328 (-2.449 to -0.339)	0.644 (0.378–0.915)
Birthweight >4500 g	Combined model	1.2%	0.830 (0.793–0.855)	0.068 (0.041–0.124)	0.011 (0.009–0.014)	-1.078 (-1.977 to -0.145)	0.703 (0.484–0.933)

Table note: Values are based on calibrated out-of-fold predicted probabilities from repeated 10-fold cross-validation. Class weights were defined using the square root of the negative-to-positive sample ratio, with the negative class assigned a weight of 1. Probability calibration was performed using isotonic regression within the cross-validation framework. Confidence intervals were obtained by bootstrap resampling. AUROC: area under the receiver operating characteristic curve; AUPRC: area under the precision–recall curve; CI: confidence interval.

Supplementary Table S6. Negative-control-type sensitivity analysis of the alcohol-related variable with gestational age at delivery.

Outcome	Model	Estimate	95% CI	Interpretation
Gestational age at delivery, weeks	Adjusted linear regression	$\beta = 0.099$ weeks	-0.046 to 0.243	No statistically significant material association
Delivery weeks <37	Adjusted logistic regression	OR = 0.92	0.73 to 1.15	No statistically significant material association
Delivery weeks <39	Adjusted logistic regression	OR = 0.90	0.76 to 1.07	No statistically significant material association
Delivery weeks ≥ 41	Adjusted logistic regression	OR = 1.27	0.81 to 1.94	Imprecise; not statistically significant

Supplementary Table S7. Comparison of included pregnancies and sequentially excluded pregnancies

Characteristic	Included analytic cohort N=6,371	Excluded: missing diet N=1,349	P value	Excluded: incomplete Visit 2/3 ultrasound N=814	P value
Maternal age, years	27.44 (5.55)	25.20 (5.76)	<0.001	25.63 (5.40)	<0.001
Non-Hispanic White	4,081 (64.1)	645 (47.8)	<0.001	496 (60.9)	0.081
Non-Hispanic Black	694 (10.9)	332 (24.6)	<0.001	87 (10.7)	0.859
Hispanic	1,022 (16.0)	242 (17.9)	0.084	171 (21.0)	<0.001
Asian/Other race or ethnicity	574 (9.0)	130 (9.6)	0.479	60 (7.4)	0.117
Income	454.99 (308.34)	359.06 (305.60)	<0.001	346.81 (274.31)	<0.001
Smoking before pregnancy	1,078 (16.9)	317 (23.5)	<0.001	132 (16.2)	0.613
Education level: $\leq$ HS grad	1,097 (17.2)	397 (29.4)	<0.001	186 (22.9)	<0.001
Education level: Some college or Assoc/Tech degree	1,778 (27.9)	470 (34.8)	<0.001	288 (35.4)	<0.001
Education level: Completed college	1,887 (29.6)	255 (18.9)	<0.001	223 (27.4)	0.190
Education level: Degree work beyond college	1,609 (25.3)	227 (16.8)	<0.001	117 (14.4)	<0.001
Family history of diabetes	1,369 (21.5)	286 (21.2)	0.815	164 (20.1)	0.379
Preexisting diabetes	88 (1.4)	28 (2.1)	0.057	15 (1.8)	0.297
Gestational diabetes	284 (4.5)	60 (4.4)	0.987	22 (2.7)	0.020
Male fetal sex	3,244 (50.9)	707 (52.4)	0.320	433 (53.2)	0.221
Visit-1 BMI, kg/m ²	26.30 (6.25)	26.96 (6.66)	0.001	26.11 (6.34)	0.440

Early gestational weight gain	2.12 (3.78)	2.08 (4.71)	0.776	2.42 (3.84)	0.040
Visit-1 waist circumference	82.66 (13.10)	84.80 (14.24)	<0.001	83.75 (13.31)	0.029
Visit-1 waist over iliac crest	94.83 (14.21)	95.82 (15.22)	0.034	94.80 (14.04)	0.958
Visit-1 hip circumference	104.13 (12.71)	105.03 (14.15)	0.038	104.36 (13.08)	0.649
Visit-1 neck circumference	32.81 (3.01)	33.20 (3.09)	<0.001	33.12 (2.90)	0.009
Visit-1 systolic blood pressure	109.05 (10.73)	109.97 (11.42)	0.008	109.28 (11.62)	0.597
Visit-1 diastolic blood pressure	67.08 (8.29)	66.36 (8.58)	0.006	67.43 (8.60)	0.278
Gestational age at delivery, weeks	38.51 (1.83)	38.19 (2.38)	<0.001	37.86 (3.26)	<0.001
Preterm birth <37 weeks	880 (13.8)	249 (18.5)	<0.001	151 (18.6)	<0.001
Birthweight, g	3,296.44 (534.50)	3,194.24 (586.57)	<0.001	3,165.25 (711.09)	<0.001
Birthweight >4000 g	505 (7.9)	82 (6.1)	0.029	44 (5.4)	0.015
Birthweight >4500 g	77 (1.2)	6 (0.4)	0.015	4 (0.5)	0.072

Supplementary Table S8. Threshold-based operating characteristics of calibrated models at the upper quintile and upper decile of predicted risk.

Outcome	Model	Threshold	Predicted-risk cutoff	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)
Intergrowth birthweight centile >97th percentile	Model 1	Upper quintile (top 20%)	9.4%	37.2	81.4	13.7	94.2
Intergrowth birthweight centile >97th percentile	Model 1	Upper decile (top 10%)	13.3%	22.1	91.0	16.3	93.6
Intergrowth birthweight centile >97th percentile	Model 2	Upper quintile (top 20%)	9.9%	42.1	81.7	15.5	94.7
Intergrowth birthweight centile >97th percentile	Model 2	Upper decile (top 10%)	14.1%	24.3	91.1	17.9	93.8
Intergrowth birthweight centile >97th percentile	Model 3	Upper quintile (top 20%)	9.9%	49.4	82.3	18.2	95.3

Intergrowth birthweight centile >97th percentile	Model 3	Upper decile (top 10%)	15.8%	34.5	91.9	25.4	94.6
Intergrowth birthweight centile >97th percentile	Combined model	Upper quintile (top 20%)	10.0%	50.6	82.4	18.7	95.4
Intergrowth birthweight centile >97th percentile	Combined model	Upper decile (top 10%)	16.7%	34.3	91.9	25.2	94.6
Intergrowth birthweight centile >90th percentile	Model 1	Upper quintile (top 20%)	26.9%	29.8	82.5	30.8	81.8
Intergrowth birthweight centile >90th percentile	Model 1	Upper decile (top 10%)	30.7%	15.2	91.3	31.3	80.5
Intergrowth birthweight centile >90th percentile	Model 2	Upper quintile (top 20%)	28.7%	33.1	83.4	34.2	82.7
Intergrowth birthweight centile >90th percentile	Model 2	Upper decile (top 10%)	34.3%	17.7	92.0	36.5	81.1
Intergrowth birthweight centile >90th percentile	Model 3	Upper quintile (top 20%)	31.5%	40.4	85.3	41.8	84.6
Intergrowth birthweight centile >90th percentile	Model 3	Upper decile (top 10%)	42.0%	24.2	93.7	50.0	82.6
Intergrowth birthweight centile >90th percentile	Combined model	Upper quintile (top 20%)	31.2%	41.8	85.7	43.3	84.9
Intergrowth birthweight centile >90th percentile	Combined model	Upper decile (top 10%)	42.8%	24.6	93.8	50.9	82.7
Birthweight >4000 g	Model 1	Upper quintile (top 20%)	12.3%	35.8	81.4	14.2	93.6
Birthweight >4000 g	Model 1	Upper decile (top 10%)	15.1%	19.0	90.8	15.0	92.9

Birthweight >4000 g	Model 2	Upper quintile (top 20%)	12.1%	39.6	81.7	15.7	94.0
Birthweight >4000 g	Model 2	Upper decile (top 10%)	15.4%	21.0	90.9	16.6	93.0
Birthweight >4000 g	Model 3	Upper quintile (top 20%)	12.7%	47.1	82.3	18.7	94.8
Birthweight >4000 g	Model 3	Upper decile (top 10%)	17.4%	29.7	91.7	23.5	93.8
Birthweight >4000 g	Combined model	Upper quintile (top 20%)	12.6%	48.3	82.4	19.1	94.9
Birthweight >4000 g	Combined model	Upper decile (top 10%)	17.9%	30.7	91.8	24.3	93.9
Birthweight >4500 g	Model 1	Upper quintile (top 20%)	2.1%	49.4	80.3	3.0	99.2
Birthweight >4500 g	Model 1	Upper decile (top 10%)	3.2%	22.1	90.1	2.7	99.0
Birthweight >4500 g	Model 2	Upper quintile (top 20%)	2.1%	55.8	80.4	3.4	99.3
Birthweight >4500 g	Model 2	Upper decile (top 10%)	3.2%	27.3	90.2	3.3	99.0
Birthweight >4500 g	Model 3	Upper quintile (top 20%)	1.7%	66.2	80.6	4.0	99.5
Birthweight >4500 g	Model 3	Upper decile (top 10%)	3.2%	46.8	90.4	5.6	99.3
Birthweight >4500 g	Combined model	Upper quintile (top 20%)	1.7%	66.2	80.6	4.0	99.5
Birthweight >4500 g	Combined model	Upper decile (top 10%)	3.3%	45.5	90.4	5.5	99.3

Table note. Threshold-based operating characteristics were calculated using calibrated out-of-fold predicted probabilities. For each outcome and model, pregnancies in the upper quintile and upper decile of predicted risk were classified as screen-positive. The predicted-risk cutoff represents the minimum calibrated predicted probability among pregnancies classified as screen-positive. Sensitivity, specificity, PPV, and NPV are presented as percentages. These thresholds represent fixed risk-stratification fractions rather than recommended clinical intervention thresholds. PPV: positive predictive value; NPV: negative predictive value.

Supplementary Table S9. Sensitivity comparison of logistic regression and non-linear classifiers for primary macrosomia outcomes.

Outcome	Model	Classifier	AUROC (95% CI)	AUPRC (95% CI)
Birthweight >4000 g	Model 1	Logistic regression	0.659 (0.639–0.674)	0.132 (0.116–0.152)
Birthweight >4000 g	Model 1	Random forest	0.646 (0.620–0.669)	0.130 (0.115–0.150)
Birthweight >4000 g	Model 1	XGBoost	0.649 (0.625–0.674)	0.132 (0.116–0.154)
Birthweight >4000 g	Model 1	Gradient boosting	0.652 (0.627–0.675)	0.130 (0.114–0.149)
Birthweight >4000 g	Model 2	Logistic regression	0.668 (0.646–0.684)	0.136 (0.121–0.158)
Birthweight >4000 g	Model 2	Random forest	0.649 (0.623–0.674)	0.139 (0.122–0.161)
Birthweight >4000 g	Model 2	XGBoost	0.663 (0.640–0.688)	0.149 (0.130–0.176)
Birthweight >4000 g	Model 2	Gradient boosting	0.652 (0.627–0.678)	0.139 (0.121–0.162)
Birthweight >4000 g	Model 3	Logistic regression	0.745 (0.725–0.768)	0.203 (0.175–0.235)
Birthweight >4000 g	Model 3	Random forest	0.732 (0.709–0.753)	0.215 (0.185–0.250)
Birthweight >4000 g	Model 3	XGBoost	0.745 (0.724–0.767)	0.212 (0.185–0.248)
Birthweight >4000 g	Model 3	Gradient boosting	0.737 (0.715–0.760)	0.205 (0.178–0.237)
Birthweight >4000 g	Combined model	Logistic regression	0.749 (0.726–0.781)	0.211 (0.190–0.236)
Birthweight >4000 g	Combined model	Random forest	0.731 (0.709–0.753)	0.208 (0.181–0.243)
Birthweight >4000 g	Combined model	XGBoost	0.737 (0.716–0.759)	0.202 (0.176–0.236)
Birthweight >4000 g	Combined model	Gradient boosting	0.731 (0.710–0.755)	0.211 (0.176–0.234)
Birthweight >4500 g	Model 1	Logistic regression	0.735 (0.688–0.777)	0.028 (0.020–0.042)
Birthweight >4500 g	Model 1	Random forest	0.698 (0.630–0.761)	0.023 (0.016–0.034)
Birthweight >4500 g	Model 1	XGBoost	0.664 (0.598–0.726)	0.019 (0.014–0.028)
Birthweight >4500 g	Model 1	Gradient boosting	0.686 (0.614–0.753)	0.020 (0.014–0.028)
Birthweight >4500 g	Model 2	Logistic regression	0.745 (0.689–0.783)	0.030 (0.021–0.036)
Birthweight >4500 g	Model 2	Random forest	0.708 (0.639–0.773)	0.025 (0.019–0.039)
Birthweight >4500 g	Model 2	XGBoost	0.675 (0.607–0.739)	0.023 (0.018–0.034)
Birthweight >4500 g	Model 2	Gradient boosting	0.712 (0.652–0.771)	0.025 (0.019–0.036)

Birthweight >4500 g	Model 3	Logistic regression	0.822 (0.790–0.862)	0.066 (0.041–0.119)
Birthweight >4500 g	Model 3	Random forest	0.803 (0.767–0.864)	0.056 (0.037–0.089)
Birthweight >4500 g	Model 3	XGBoost	0.775 (0.712–0.836)	0.062 (0.040–0.107)
Birthweight >4500 g	Model 3	Gradient boosting	0.798 (0.741–0.852)	0.065 (0.041–0.109)
Birthweight >4500 g	Combined model	Logistic regression	0.830 (0.794–0.855)	0.068 (0.041–0.124)
Birthweight >4500 g	Combined model	Random forest	0.821 (0.774–0.866)	0.057 (0.037–0.091)
Birthweight >4500 g	Combined model	XGBoost	0.784 (0.726–0.839)	0.063 (0.041–0.104)
Birthweight >4500 g	Combined model	Gradient boosting	0.817 (0.773–0.859)	0.060 (0.038–0.101)

Table note. Values are AUROC or AUPRC with 95% confidence intervals in parentheses. Logistic-regression rows correspond to the primary class-weighted, isotonic-calibrated models. Non-linear classifiers were included as supplementary sensitivity analyses; their hyperparameters were selected within each outer training fold using an inner 3-fold randomized search optimizing average precision. Class imbalance was handled using fold-specific sample weights, with the negative class assigned a weight of 1 and the positive class weighted by the square root of the negative-to-positive sample ratio. Confidence intervals were estimated using 2,000 bootstrap resamples of out-of-fold predictions. AUROC: area under the receiver operating characteristic curve; AUPRC: area under the precision-recall curve.

Supplementary Table S10. Missingness of retained candidate predictors before k-nearest-neighbor imputation.

Risk factors	Missing, %
Age	0.4
BMI	1.3
Non-Hispanic white	0.0
Non-Hispanic black	0.0
Hispanic	0.0
Asian and other	0.0
Family history of diabetes	0.0
Income level	16.3
Smoke in 3 months before pregnancy	0.0
Fetal sex	0.0
Education level	0.0
weight before pregnancy	2.0
Weight gain at visit 1	1.6
Weight gain at visit 2	2.2
Weight gain at visit 3	2.1
Waist at visit 1	2.0
Waist over iliac crest at visit 1	2.0
Hip circumference at visit 1	2.1
Neck circumference at visit 1	12.1
Systolic BP at visit 1	1.4
Diastolic BP at visit 1	1.4
Systolic BP at visit 2	1.4
Diastolic BP at visit 2	1.4
Systolic BP at visit 3	1.3
Diastolic BP at visit 3	1.3

Diabetes	0.0
GDM	0.0
WK Gestational age by project EDD on U/S date at visit 2	0.5
U2AB02 Biparietal Diameter at visit 2	0.0
U2AB03 Head Circumference at visit 2	0.0
U2AB04 Abdominal Circumference at visit 2	0.2
U2AB05 Femur Diaphysis Length at visit 2	0.0
U2AB06 Estimated fetal weight as reported at visit 2	0.2
U2CC06 Left uterine artery - Resistance Index (RI) at visit 2	9.2
U2CC07 Left uterine artery - Pulsatility Index (PI) at visit 2	9.3
U2CC08 Left uterine artery - Systolic acceleration time (milliseconds) at visit 2	10.2
Right uterine artery - Heart rate (bpm) at visit 2	9.4
Right uterine artery - Angle of insonation (degrees) at visit 2	9.6
Right uterine artery - Peak Systolic Velocity (PSV) (cm/sec) at visit 2	9.0
Right uterine artery - Systolic/Diastolic ratio (S/D) at visit 2	9.4
Right uterine artery - Resistance Index (RI) at visit 2	9.0
Right uterine artery - Pulsatility Index (PI) at visit 2	9.2
Right uterine artery - Systolic acceleration time (milliseconds) at visit 2	10.0
Gestational age by project EDD on U/S date at visit 3	0.4
Biparietal diameter at visit 3	0.0
Head circumference at visit 3	0.0
Abdominal circumference at visit 3	0.0
Femur diaphysis length at visit 3	0.0
Amniotic Fluid Index - Quadrant 1 at visit 3	7.4
Amniotic Fluid Index - Quadrant 2 at visit 3	7.2
Amniotic Fluid Index - Quadrant 3 at visit 3	7.1
Amniotic Fluid Index - Quadrant 4 at visit 3	8.0
Estimated fetal weight as reported-grams (from nuMoM2b)	0.0
Estimated fetal weight as reported – Percentile (from nuMoM2b)	0.5
Cervical length	2.4
HEI / AHEI (diet)	0.0

Note: Missingness was calculated in the final analytic cohort before k-nearest-neighbor imputation. The denominator for all percentages was N=6,371. The table includes retained candidate predictors used in the model-development framework. Ultrasound availability during cohort derivation was defined primarily according to Visit 2 and Visit 3 fetal biometry measurements. Other ultrasound-related variables were evaluated during predictor selection, and variables with substantial missingness were not retained as candidate predictors. Residual missingness among retained predictors was handled using k-nearest-neighbor imputation within the training data of each cross-validation split. Percentages are rounded to one decimal place.

Supplementary Table S11. Comparison of WHO and INTERGROWTH-21st fetal ultrasound percentile comparators for primary macrosomia outcomes.

Outcome	Visit	Measure	WHO AUROC	INTERGROWTH-21st AUROC (95% CI)
Birthweight >4000 g	Visit 2	BPD	0.570	0.572 (0.547–0.599)
Birthweight >4000 g	Visit 2	HC	0.580	0.581 (0.558–0.605)
Birthweight >4000 g	Visit 2	AC	0.590	0.589 (0.564–0.614)
Birthweight >4000 g	Visit 2	FL	0.540	0.548 (0.523–0.574)

Birthweight >4000 g	Visit 3	BPD	0.650	0.660 (0.634–0.685)
Birthweight >4000 g	Visit 3	HC	0.650	0.664 (0.638–0.688)
Birthweight >4000 g	Visit 3	AC	0.700	0.707 (0.685–0.730)
Birthweight >4000 g	Visit 3	FL	0.610	0.610 (0.584–0.636)
Birthweight >4000 g	Visit 3	EFW	0.680	0.694 (0.669–0.717)
Birthweight >4500 g	Visit 2	BPD	0.580	0.578 (0.513–0.646)
Birthweight >4500 g	Visit 2	HC	0.620	0.619 (0.548–0.683)
Birthweight >4500 g	Visit 2	AC	0.640	0.639 (0.581–0.696)
Birthweight >4500 g	Visit 2	FL	0.540	0.551 (0.481–0.618)
Birthweight >4500 g	Visit 3	BPD	0.740	0.752 (0.694–0.805)
Birthweight >4500 g	Visit 3	HC	0.740	0.747 (0.685–0.806)
Birthweight >4500 g	Visit 3	AC	0.790	0.799 (0.745–0.846)
Birthweight >4500 g	Visit 3	FL	0.680	0.690 (0.628–0.749)
Birthweight >4500 g	Visit 3	EFW	0.770	0.787 (0.734–0.834)

Note. Fetal ultrasound percentiles were evaluated as single-predictor chart-based comparators for subsequent macrosomia. INTERGROWTH-21st values are shown as AUROC with 95% confidence intervals from 2,000 bootstrap resamples. AUROC: area under the receiver operating characteristic curve; WHO: World Health Organization; BPD: biparietal diameter; HC: head circumference; AC: abdominal circumference; FL: femur length; EFW: estimated fetal weight.

Supplementary Table S12. TRIPOD+AI 2024 checklist.

Item number	Checklist item	Reported in manuscript	Location	Comments
1	Title identifies the study as developing and internally validating a prediction model, the target population, and outcome.	Yes	Title page	The title states development and internal validation of prediction models for macrosomia in nulliparous singleton pregnancies.
2	Abstract summarizes objectives, data source, participants, predictors, outcomes, analytical methods, results, and conclusions.	Yes	Abstract	The abstract reports the nuMoM2b cohort, sample size, predictors, outcomes, logistic regression models, internal validation, comparison with fetal growth chart-based benchmarks,

				and main performance results.
3a	Background explains the clinical context and rationale for prediction modeling.	Yes	Introduction	The introduction describes macrosomia-related clinical risks and the need for earlier risk stratification.
3b	Target population, intended purpose, and clinical context of model use are described.	Yes	Introduction; Discussion, Sections 4.1 and 4.3	The intended context is antenatal risk stratification for macrosomia in nulliparous singleton pregnancies.
3c	Relevant population groups or sociodemographic considerations are described where applicable.	Partly	Section 2.3; Table 1; Supplementary Table S7; Discussion, Section 4.4	Sociodemographic predictors such as race/ethnicity and income were included and summarized. Potential selection during cohort derivation was evaluated in Supplementary Table S7. Formal fairness or subgroup performance analyses were not performed.
4	Study objectives specify model development and validation purpose.	Yes	End of Introduction	The study objective states development and internal validation of visit-updated prediction models for birthweight >4000 g and >4500 g.
5a	Data source and rationale are described.	Yes	Section 2.1	The study is a secondary analysis of the prospective nuMoM2b cohort.
5b	Key dates of data collection are reported.	Yes	Section 2.1	Recruitment period and scheduled visit windows are described.
6a	Study setting is described.	Yes	Section 2.1	Participants were recruited from hospitals affiliated with eight clinical centers in the United States.
6b	Eligibility criteria and participant selection are described.	Yes	Section 2.1; Section 3.1; Supplementary Figure S4	Inclusion and exclusion criteria are described, including availability of periconceptional diet, Visit 2 and Visit 3 ultrasound data, birth outcome data, and live-birth status. Participant flow is shown in Supplementary Figure S4.
6c	Treatments or interventions are described if relevant.	Not applicable	—	This was an observational secondary analysis; no treatment assignment or intervention was part of the prediction model study.
7	Data preparation, preprocessing, and quality checking are described.	Yes	Sections 2.3–2.5; Supplementary Methods; Supplementary	Predictor construction, ultrasound-derived percentile variables, missing-data handling, imputation, scaling, and cross-validation

			Tables S3 and S10	preprocessing are described. WHO fetal growth chart–based percentile variables are defined in Supplementary Table S3, and predictor-level missingness is reported in Supplementary Table S10.
8a	Outcome definition, time horizon, assessment, and rationale are described.	Yes	Section 2.2; Section 3.1; Supplementary Tables S4	Primary outcomes were birthweight >4000 g and >4500 g. LGA90 and LGA97 were used as sensitivity outcomes and are reported in Supplementary Tables S4.
8b	Qualifications of outcome assessors are described if subjective assessment is required.	Not applicable	Section 2.2	Outcomes were based on recorded birthweight, gestational age at delivery, and neonatal sex, not subjective adjudication.
8c	Blinding of outcome assessment is described if relevant.	Not applicable	—	Outcome assessment was based on recorded clinical birth data; no subjective blinded assessment was required.
9a	Initial predictor selection and rationale are described.	Yes	Section 2.3	Candidate predictors were prespecified according to clinical availability during antenatal care and prior evidence linking them to fetal overgrowth.
9b	Predictors are clearly defined, including how and when they were measured.	Yes	Section 2.3; Table 1; Supplementary Methods; Supplementary Table S3	Maternal characteristics, anthropometry, diet indices, fetal sex, and ultrasound predictors are described in the Methods and Table 1. WHO fetal growth chart–based percentile variables are further defined in Supplementary Table S3.
9c	Qualifications of predictor assessors are described if subjective interpretation is required.	Partly	Section 2.1; Section 2.3	Most predictors were routinely collected clinical, anthropometric, ultrasound, or questionnaire variables. Ultrasound measurements followed the standardized nuMoM2b visit structure.
10	Sample size is explained and justified in the prediction-model context.	Yes	Section 2.5; Section 3.1	The revised manuscript states that sample size was determined by eligible nuMoM2b data after prespecified exclusions and reports the final analytic sample and event counts for birthweight >4000 g and >4500 g.

11	Missing data handling is described, including reasons for omitting data.	Yes	Sections 2.1 and 2.5; Supplementary Figure S4; Supplementary Tables S7 and S10	Cohort-level exclusions are shown in Supplementary Figure S4. Included versus excluded pregnancies are compared in Supplementary Table S7. Predictor-level missingness before imputation is reported in Supplementary Table S10, and k-nearest-neighbor imputation is described in Section 2.5.
12a	Data use for model development and internal validation is described.	Yes	Section 2.5	Repeated stratified 10-fold cross-validation with 1000 repetitions was used for internal validation.
12b	Predictor handling, transformation, scaling, or standardization is described.	Yes	Section 2.5	Imputation and scaling were performed within the training data only to avoid information leakage.
12c	Model type, rationale, model-building steps, hyperparameter tuning, and internal validation are described.	Yes	Section 2.5; Supplementary Table S9	Logistic regression was selected as the primary modeling approach. Regularization, class weighting, hyperparameter tuning, isotonic calibration, and internal validation are described in Section 2.5. Non-linear classifier sensitivity comparisons are reported in Supplementary Table S9.
12d	Handling of heterogeneity across clusters or centers is described if performed.	Not performed	—	Formal center- or cluster-specific model-performance heterogeneity was not evaluated.
12e	Performance measures and plots are specified.	Yes	Section 2.5; Section 3.2; Figure 1; Supplementary Figures S1–S3; Supplementary Tables S4–S6 and S8–S9	AUROC, AUPRC, Brier score, calibration intercept, calibration slope, reliability plots, LGA sensitivity performance, threshold-based operating characteristics, and non-linear model sensitivity comparisons are reported.
12f	Model updating is described if performed.	Not applicable	—	No external model updating, recalibration, or refitting after external validation was performed.
12g	Calculation of model predictions for evaluation is described.	Yes	Section 2.5; Supplementary Tables S5–S6 and S9	Out-of-fold predicted probabilities were generated within the cross-validation framework, calibrated using isotonic regression, and used for discrimination,

				calibration, and threshold-based operating-characteristic analyses.
13	Class imbalance handling and recalibration are described.	Yes	Section 2.5; Supplementary Tables S4-S5; Supplementary Figures S2-S3	Class-weighted logistic regression was used, followed by isotonic calibration. Calibration summaries are reported for primary outcomes and LGA sensitivity outcomes.
14	Model fairness approaches are described if performed.	Not performed	Section 2.3; Table 1; Supplementary Table S7; Discussion, Section 4.4	Race/ethnicity and income were included as predictors and summarized. Selection during cohort derivation was assessed in Supplementary Table S7. Formal fairness or subgroup performance analyses were not performed.
15	Model output and thresholds are described.	Yes	Section 2.5; Section 3.2; Supplementary Table S8	Model output was calibrated predicted probability. Upper-quintile and upper-decile risk strata were evaluated as prespecified risk-stratification thresholds in Supplementary Table S8.
16	Differences between development and evaluation data are identified.	Not applicable	Section 2.5	The study used internal validation within the same cohort; no separate external evaluation dataset was used.
17	Ethical approval and informed consent are reported.	Yes	Ethics Statement	Original nuMoM2b informed consent and institutional approvals for this secondary analysis are reported.
18a	Funding source and funder role are reported.	Yes	Funding Information	Funding source and funder role are stated.
18b	Conflicts of interest are reported.	Yes	Conflict of Interest Statement	Authors declare no conflicts of interest.
18c	Protocol availability is stated.	Partly	Ethics Statement; Data Availability Statement	The secondary analysis was approved by relevant ethics boards. A separate public analysis protocol was not reported.
18d	Study registration is reported if applicable.	Not applicable	—	This secondary analysis was not registered as a clinical trial.
18e	Data availability is reported.	Yes	Data Availability Statement	nuMoM2b data are available from the NICHD DASH repository upon application and approval.
18f	Analytical code availability is reported.	Partly	Data Availability Statement; Methods	Analytical procedures are described in the Methods. Public code release was not specified.

19	Patient and public involvement is reported if applicable.	Not applicable	—	No patient or public involvement was included in the design, conduct, reporting, or dissemination of this secondary analysis.
20a	Participant flow is described, including number with and without outcome.	Yes	Section 3.1; Supplementary Figure S4	Participant flow and sequential exclusions are reported in Section 3.1 and Supplementary Figure S4.
20b	Participant characteristics and missing data are summarized.	Yes	Table 1; Supplementary Tables S1–S2, S7, and S10	Baseline characteristics are shown in Table 1. Univariate associations are provided in Supplementary Tables S1 and S2. Included versus excluded pregnancies are compared in Supplementary Table S7. Predictor-level missingness is reported in Supplementary Table S10.
20c	Distribution of predictors and outcomes is compared between development and evaluation data if applicable.	Not applicable	—	No separate external evaluation dataset was used.
21	Number of participants and outcome events in each analysis is specified.	Yes	Sections 2.5 and 3.1; Supplementary Tables S4–S6 and S9	The final analytic sample and event counts for primary and LGA sensitivity outcomes are reported. Event rates are also shown in model-performance and threshold-based operating-characteristic tables.
22	Full model specification or availability is provided to support future evaluation or implementation.	Partly	Section 2.5; Figures 2–3	Modeling procedures and feature contributions are reported. The study does not present a finalized deployable clinical model object.
23a	Model performance estimates are reported with confidence intervals and relevant plots.	Yes	Section 3.2; Figure 1; Table 2; Supplementary Figures S1–S3; Supplementary Tables S4–S6, S9–S10, and S11	Discrimination, precision–recall performance, calibration, threshold operating characteristics, non-linear classifier sensitivity comparisons, and WHO versus INTERGROWTH-21st chart-comparator analyses are reported.
23b	Heterogeneity in model performance across clusters is reported if examined.	Not performed	—	Center- or cluster-specific performance heterogeneity was not evaluated.
24	Model updating results are reported if performed.	Not applicable	—	No model updating after external validation was performed.

25	Overall interpretation of results is provided in context of objectives and previous studies.	Yes	Discussion, Sections 4.1–4.3	The Discussion interprets the visit-updated prediction pattern, comparison with fetal growth chart–based benchmarks, dietary findings, clinical implications, and relationship to previous studies.
26	Limitations are discussed, including sample size, missing data, overfitting, representativeness, and generalizability.	Yes	Discussion, Section 4.4; Supplementary Tables S7 and S10	Limitations include internal validation only, selection bias, rare >4500 g outcome, missingness, ultrasound timing, Hadlock-derived EFW limitations, dietary recall bias, and need for external validation. Selection and missingness are further summarized in Supplementary Tables S7 and S10.
27a	Handling of unavailable or poor-quality input data in future implementation is discussed.	Partly	Section 2.5; Discussion, Sections 4.3–4.4; Supplementary Table S10	Missing data were handled during model development using k-nearest-neighbor imputation within cross-validation. Future implementation would require prospective assessment of input-data availability and quality.
27b	Required user interaction and expertise for model use are discussed.	Partly	Discussion, Section 4.3	The model is positioned as a potential adjunct for antenatal risk stratification rather than a replacement for clinical assessment. A specific implementation workflow was not evaluated.
27c	Next steps for future research, applicability, and generalizability are discussed.	Yes	Discussion, Section 4.4; Conclusion	External validation and further evaluation before clinical implementation are explicitly stated.

Note. TRIPOD+AI: Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis plus Artificial Intelligence. This checklist summarizes reporting coverage for the present model-development and internal-validation study. Items relating to external validation, model updating, deployment, or clinical implementation were marked as not applicable or partly reported where these analyses were not performed.

Supplementary Table S13. Limited external validation of the reduced early-pregnancy model in an independent Dutch cohort.

Outcome	Model / evaluation setting	Predictor set	AUROC
Birthweight >4000 g	Original nuMoM2b Visit-1 model, internal validation	Full Visit-1 predictor set including periconceptional diet	0.66
Birthweight >4000 g	Reduced nuMoM2b early-pregnancy model, internal validation	Harmonizable maternal demographic and clinical predictors only	0.64

Birthweight >4000 g	Reduced nuMoM2b early-pregnancy model, external validation in MMC	Harmonizable maternal demographic and clinical predictors only	0.60
Birthweight >4000 g	Comparable MMC model, internal validation	Harmonizable maternal demographic and clinical predictors only	0.61
Birthweight >4500 g	Original nuMoM2b Visit-1 model, internal validation	Full Visit-1 predictor set including periconceptional diet	0.74
Birthweight >4500 g	Reduced nuMoM2b early-pregnancy model, internal validation	Harmonizable maternal demographic and clinical predictors only	0.67
Birthweight >4500 g	Reduced nuMoM2b early-pregnancy model, external validation in MMC	Harmonizable maternal demographic and clinical predictors only	0.64
Birthweight >4500 g	Comparable MMC model, internal validation	Harmonizable maternal demographic and clinical predictors only	0.63

Table note. This supplementary analysis evaluated a reduced early-pregnancy model using predictors harmonizable between the nuMoM2b cohort and an independent Dutch cohort from Máxima Medical Center. Harmonizable predictors included maternal age, body mass index, ethnicity/race-related background where available, and gravidity-related information. Periconceptional dietary indices and serial Visit 2/Visit 3 ultrasound biometry were not available in a comparable form in the MMC cohort and were therefore not included in this reduced external-validation analysis. The analysis should be interpreted as a limited evaluation of transportability for the shared early-pregnancy predictor component, not as external validation of the full diet- and ultrasound-augmented visit-updated models. AUROC: area under the receiver operating characteristic curve; MMC: Máxima Medical Center.